# Conditional Probability

**Exam code:** 0607

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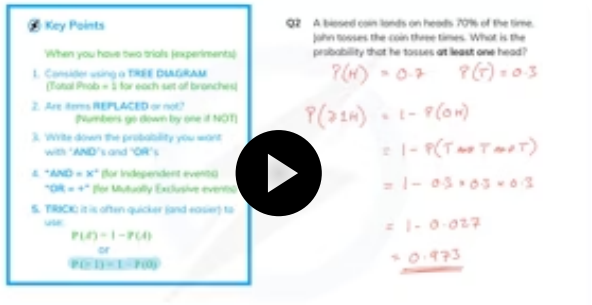


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# Conditional Probability

## What is a conditional probability?

Help me understand

- A **conditional probability** is the probability of something happening (*A*) **given that** something else has **already happened** (*B*)

### Examiner Tips and Tricks

Conditional probability can be written using formal notation.

The notation $P(A \mid B)$ describes the probability of event A occurring, **given** that event B has already occurred.

Note that you are **not** expected to know this notation for the exam, but you are expected to be able to calculate conditional probabilities using Venn diagrams, tree diagrams or tables.

**Stuck on something?**

Ask questions and get help with exactly what you're revising.

Got it

## How do I calculate conditional probabilities?

from 1, 2, 3, 4, 5, 6, 7, 8, 9

- P(it selects a multiple of three) = $\frac{3}{9}$
- This is **not** a conditional probability
- There are 3 possibilities (3, 6, 9) out of **all** 9 possibilities

- However, if you program the computer to only select from even numbers, then
  - P(it selects a multiple of three **given that** it selects an even number) = $\frac{1}{4}$
  - 1 possibility (6) out of **only** 4 possibilities (2, 4, 6, 8)
  - This **is** a conditional probability

### Examiner Tips and Tricks

- Look out for conditional probability questions within larger questions on two-way tables, Venn diagrams or tree diagrams
  - They often use the phrase **given that**

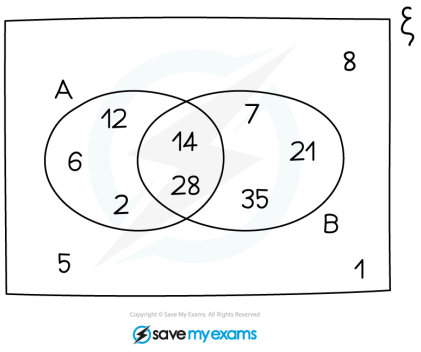
### Worked Example

A Venn diagram is shown below.

$A = \{2, 6, 12, 14, 28\}$
$B = \{7, 14, 21, 28, 35\}$
$\xi = \{1, 2, 5, 6, 7, 12, 14, 21, 28, 35\}$

A number is picked at random.

Find the probability that the number is in set $A$ given that it is in set $B$.




**Answer:**

Given that B has already happened, your probability will be out of number of numbers in $B$

$\frac{2}{5}$

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